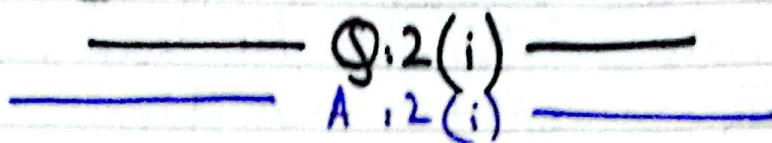


Section - B



✓ Empirical formula :-

The empirical formula of a compound tells you which is the chemical formula that gives the simplest integer ratio.

FOR EXAMPLE :- the compound of hydrogen peroxide has one H atoms for ~~4~~ every ~~8~~ every oxygen atoms. The simplest of hydrogen peroxide is 1:1. The empirical formula is written as HO.

✓ Molecular formula :-

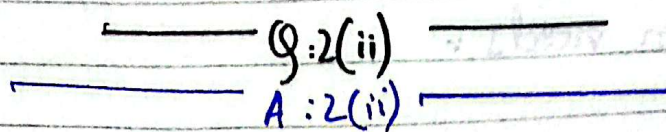
Molecular formula is the expression that specifies the actual number of atoms in each element on molecule of the compound. This shows the actual number of atoms of each element.

FOR EXAMPLE :-

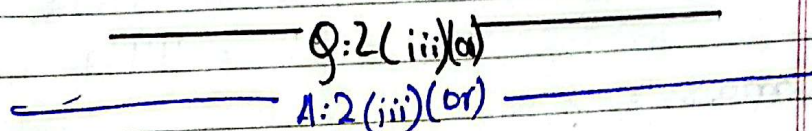
The molecular formula of hydrogen peroxide is H_2O_2 but the empirical formula of hydrogen peroxide is HO.

Day: _____

Date: _____

Molecular mass of $\text{C}_6\text{H}_{12}\text{O}_6$:

$$\begin{aligned} &= 6(12) + 12(1.008) + 6(16) \\ &= 72 + 12.096 + 96 \\ &= 84.096 + 96 \\ &= 180.096 \text{ amu.} \end{aligned}$$

Empirical
formula
MassMolecular
formula

Definition

* Empirical formula is of the compounds that gives the simplest integer ratios

Molecular formula is the expression that gives the actual number of atoms.

Other names

* Experimental formula
* Simplest formula

Calculated formula

Provide information

* It provides more information

It provides less information.

Example

* Hydrogen peroxide (H_2O_2) : 1:1

Hydrogen peroxide (H_2O_2)

Day: _____

Date: _____

Their needs :-

★

Molecular mass.

★

formula mass.

Q:2(iv)

A:2(iv)

Moles :- 0.25 mol

$$\begin{aligned} \text{molar mass of } \text{CO}_2 &= 1(12) + 2(16) \\ &= 12 + 32 \\ &= 44 \text{ g} \end{aligned}$$

mass = ?

$$\begin{aligned} \text{mass} &= \text{moles} \times \text{m.m (molar mass)} \\ &= 0.25 \times 44 \text{ g} \\ &= 11 \text{ g/mol} \end{aligned}$$

Q:2(v)

A:2(v)

moles :- 0.5

$$\begin{aligned} \text{molecules} &= \text{moles} \times N_A \text{ (avogadro's no)} \\ &= 0.5 \times 6.022 \times 10^{23} \\ &= 3.011 \times 10^{23} \text{ molecules of methane.} \end{aligned}$$

Q:2(vi)

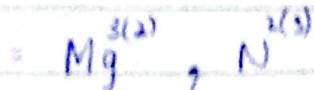
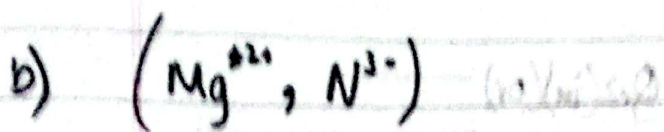
A:2(vi)

a)

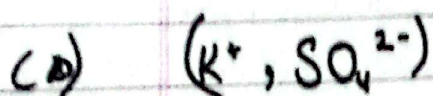
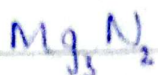
 $(\text{Na}^+, \text{Cl}^-)$ Chemical formula :- $\text{Na}^{+1}, \text{Cl}^{-1}$

1:1

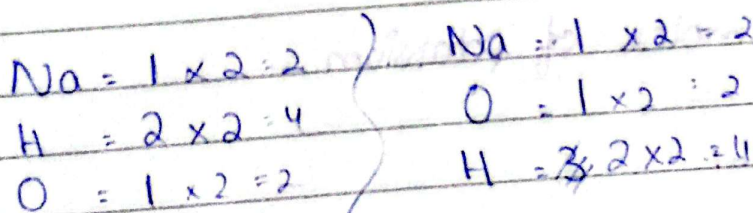
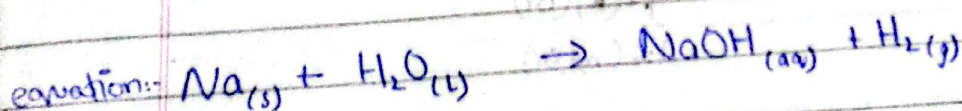
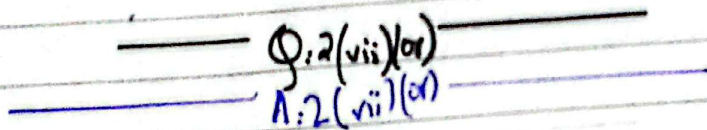
= NaCl



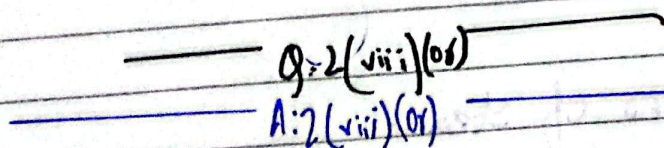
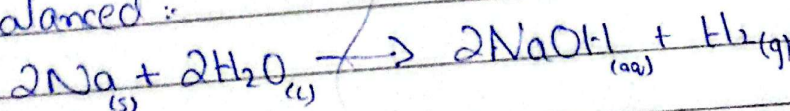
$$3 \times 2 = 6, 2 \times 3 = 6$$



$$2 \times 1 = 2, 1 \times 2 = 2$$



Balanced:



Moles = 0.2

~~molar mass of Al~~

number of atoms = $0.2 \times 6.022 \times 10^{23}$
 $= 1.2044 \times 10^{23}$

Day: _____

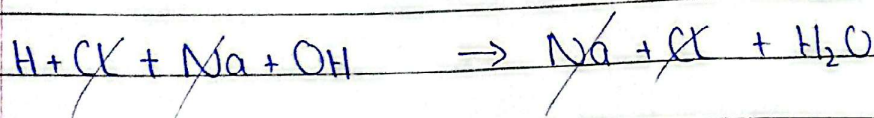
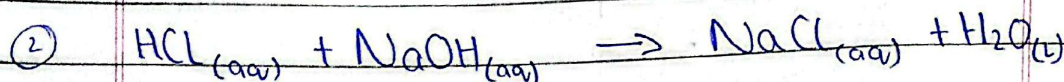
Date: _____

Q: 2 (in/ or)

A: 2 (in/ or)

Q Ionic equation:-

Ionic equation is a chemical equation that dissolved substance in water by the individual ions and spectator ions (common on both sides) are removed by net ionic ~~equation~~ equation.



Q: 2 (x/ or)

A: 2 (x/ or)

a) 1.2 moles of potassium.

moles:- 1.2

mass = ?

molar mass:- 39

mass = 1.2×39

= 46.8 g

b) 0.25 moles of steam

moles:- 0.25

mass = ?

molar mass = $2(1.008) + 16 = 2.016 + 16$

= 18.016

$$\begin{aligned} \text{mass} &= 0.25 \times 18.016 \\ &= 4.504 \text{ g} \end{aligned}$$

$$\frac{Q:2(ni)(or)}{A:2(ni)(or)}$$

$$\frac{Q:2(ni)(or)}{A:2(ni)(or)}$$

① Empirical formula of $C_8H_{10}N_4O_2$:-

$$\begin{aligned} C_8H_{10}N_4O_2 &\left\{ \begin{array}{l} 8:10:4:2 \\ 4:5:2:1 \end{array} \right\} \\ = C_4H_5N_2O &\end{aligned}$$

② Molecular mass of $C_8H_{10}N_4O_2$:-

$$\begin{aligned} &= 8(12) + 10(1) + 4(14) + 2(16) \\ &= 96 + 10 + 56 + 32 \\ &= 194 \end{aligned}$$

SECTION - C

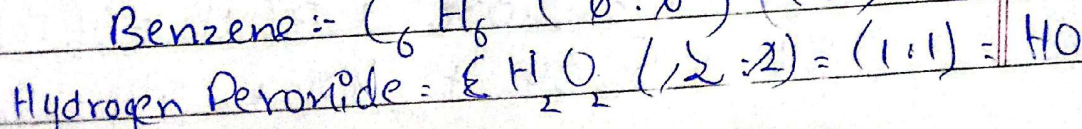
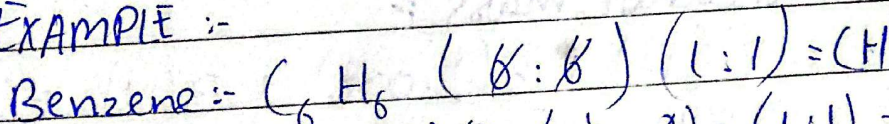
$$\frac{Q:3(i)(or)}{A:3(i)(or)}$$

$$\frac{Q:3(i)(or)}{A:3(i)(or)}$$

Empirical formula :-

Empirical formula is of the compound that gives the simplest integer ratio of an atom of each element in one molecule of the compound.

For EXAMPLE :-



Molecular formula:-
 molecular formula is the expression that specifies the actual number of atoms of each element in one molecule of the compound. This shows the actual number of atoms in each element.

FOR EXAMPLE:-

Benzene :- C_6H_6

Hydrogen Peroxide :- H_2O_2

b) i) Aspirin ($C_9H_8O_4$) :-

Empirical formula :- $C_9H_8O_4$

Molecular mass :-

$$\begin{aligned}
 &= 9(12) + 8(1.008) + 4(16) \\
 &= 108 + 8.064 + 64 \\
 &= 180.064 \text{ amu.}
 \end{aligned}$$

ii) Vinegar (CH_3COOH) :-

Empirical formula :- CH_3COOH

Molecular mass :-

$$\begin{aligned}
 &= 1(12) + 3(1.008) + 1(12) + 1(16) + 1(16) + 1(1) \\
 &= 12 + 3.024 + 12 + 16 + 16 + 1 \\
 &= 60.024 \text{ amu.}
 \end{aligned}$$

iii) TNT ($C_7H_5N_3O_6$) :-

Empirical formula :- $C_7H_5N_3O_6$

Molecular mass :-

$$\begin{aligned}
 &= 7(12) + 5(1.008) + 3(14) + 6(16) \\
 &\Rightarrow 84 + 5.04 + 42 + 96 = 227.04 \text{ amu}
 \end{aligned}$$

Q: 3(ii)(or) ———
 A: 3(ii)(or) ———

a i) 2.4g of He

moles :- ?

mass :- 2.4g

molar mass :- 4g

mole :- $\frac{\text{mass}}{\text{molar mass}}$

$$= \frac{2.4}{4}$$

$$= 0.6 \text{ mol}$$

a ii) 250mg of carbon

mass :- 0.25g

molar mass :- 14

$$\text{mole} :- \frac{0.25}{14} = 0.0178 \text{ mol}$$

$$\left(\begin{array}{l} 250 \text{ mg} \rightarrow \text{g} \\ 250 \div 1000 = 0.25 \text{ g} \end{array} \right)$$

a iii) 40g of sulphur :-

moles :- ?

mass :- 40g

molar mass :- 32

$$\text{mole} :- \frac{40}{32} = 1.25 \text{ mol}$$

Day: _____

Date: _____

b) i) 75 moles of H_2

mole :- 75

mass :- ?

$$\text{molar mass} :- 2(1.008) \\ = 2.016$$

$$\text{mass} = \text{mole} \times \text{molar mass} \\ = 75 \times 2.016 \\ = 151.2 \text{ g}$$

b) ii) 0.15 moles of H_2SO_4

mass :- ?

mole :- 0.15

$$\text{molar mass} :- 2(1.008) + 1(32) + 4(16) \\ = 2.016 + 32 + 64 \\ = 98.016$$

$$\text{mass} = \text{mole} \times \text{molar mass} \\ = 0.15 \times 98.016 \\ = 14.7024 \text{ g}$$

Q: 3. (iii) (or) _____

A: 3 (iii) (or) _____

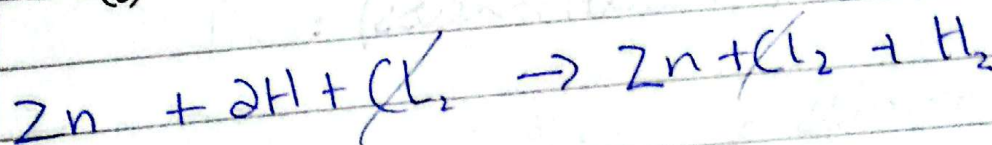
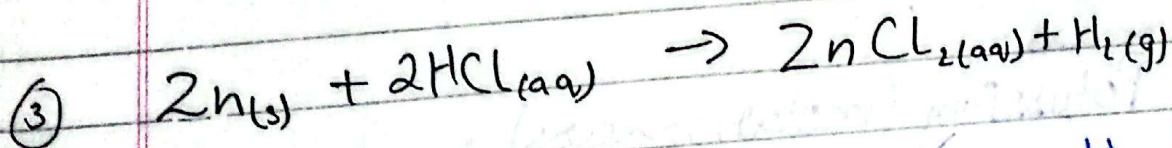
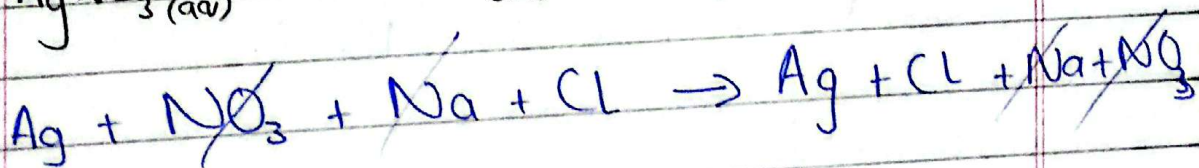
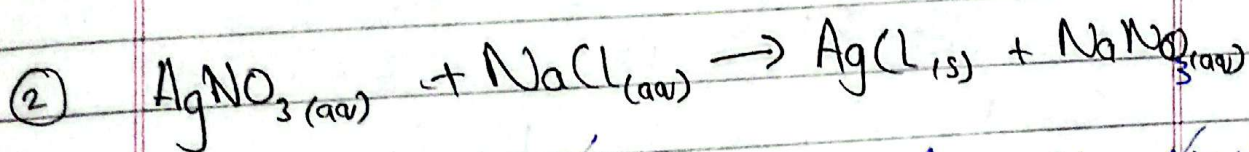
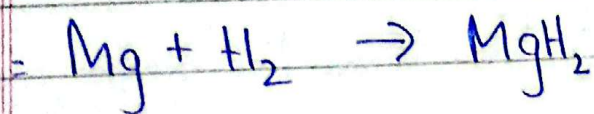
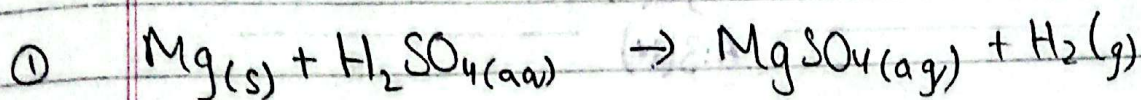
* Ionic Equation :-

Ionic equation is the chemical equation that dissolved sub. substance in water by the individual ions and spectator ions which are removed

Day: _____

Date: _____

by net ionic equation.

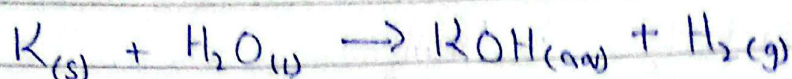


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Date: _____

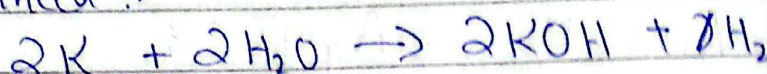
Q.3(iv)A.3(iv)

(i) Balance the equation:-



$$\left. \begin{array}{l} K = 1 \times 2 = 2 \\ H = 2 \times 2 = 4 \\ O = 1 \times 2 \end{array} \right\} \begin{array}{l} K = 1 \times 2 = 2 \\ H = 3 \times 2 = 6 \quad (2 \times 2 = 4) \\ O = 1 \times 2 \end{array}$$

Balanced :-



(ii)

Moles = ?

mass = 0.5g

Potassium (molar mass) : 39

$$\text{moles} = \frac{0.5g}{39 \text{ mol}}$$

$$= 0.0128 \text{ mol/g}$$

(iii)

number of atoms = ?

molar mass : 39

NA : 6.022×10^{23} atoms : $39 \times 6.022 \times 10^{23}$

$$= 234.858 \times 10^{23}$$

$$= 2.34 \times 10^2 \times 10^{23} \rightarrow 2.34 \times 10^{25}$$

BOARD OF INTERMEDIATE AND SECONDARY EDUCATION

Annual Examination 2026

Chemistry — SSC-I (9th Grade)

Syllabus: Chapter 6 — Stoichiometry

Total Marks: 46**Time Allowed: 1h 45m****Version: 1****General Instructions:**

1. Write your name, roll number and section clearly on the answer sheet.
2. Section A is compulsory. Encircle the correct option on the question paper itself.
3. In Section B, attempt any **8** questions out of 11. For each question, attempt either the main question **OR** its alternative.
4. In Section C, attempt any **2** questions out of 4. For each question, attempt either the main question **OR** its alternative.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The empirical formula of glucose ($C_6H_{12}O_6$) is:	$C_6H_{12}O_6$	CH_2O	$C_2H_4O_2$	CHO	A●C●D
2	The molecular mass of naphthalene ($C_{10}H_8$) is:	120 amu	108 amu	128 amu	136 amu	A●B●D
3	The formula mass of $Mg(OH)_2$ is: (Mg=24, O=16, H=1)	40 amu	56 amu	58 amu	74 amu	A●C●D
4	One mole of any substance contains how many representative particles?	6.022×10^{22}	6.022×10^{23}	6.022×10^{24}	6.022×10^{20}	A●C●D
5	How many moles of molecules are there in 16 g of oxygen gas (O_2)?	1 mol	0.5 mol	0.1 mol	2 mol	A●C●D
6	The mass of 9.05 moles of ozone (O_3) is: (O=16)	48 g	144 g	434.4 g	480 g	A●B●D
7	How many atoms are present in 1.25 moles of zinc (Zn)?	6.022×10^{23}	7.53×10^{23}	4.82×10^{23}	1.25×10^{23}	A●C●D
8	The formula of aluminium oxide (Al^{3+} , O^{2-}) is:	AlO	Al_2O_3	Al_3O_2	AlO_3	A●C●D
9	Which compound has the same empirical and molecular formula?	H_2O_2	C_6H_6	H_2O	$C_6H_{12}O_6$	A●B●D
10	What is the mass of 4 moles of hydrogen gas H_2 ? (H=1.008)	8.064 g	4.032 g	1 g	1.008 g	●B●C●D
11	In a chemical equation, balancing must be done by changing:	subscripts	symbols	coefficients	state symbols	A●B●D
12	In an ionic equation, ions that appear on both sides and do not participate are called:	reactive ions	spectator ions	anions	cations	A●C●D

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt any 8 questions. For each pair, attempt either the main question OR its alternative.